



Biochip Course

Biosensors: Cells

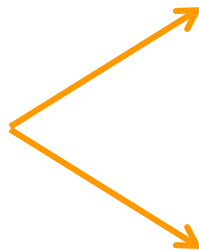
Marco Carminati

a.y. 2022/2023

April 17th 2023

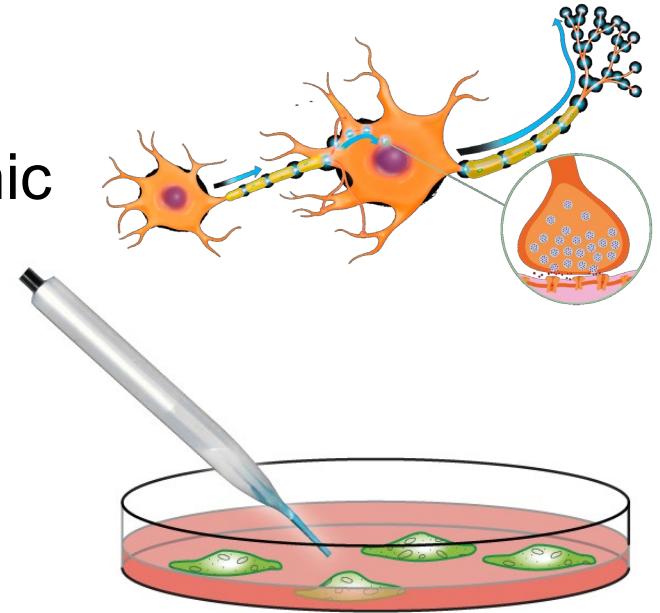


Interface with Cells



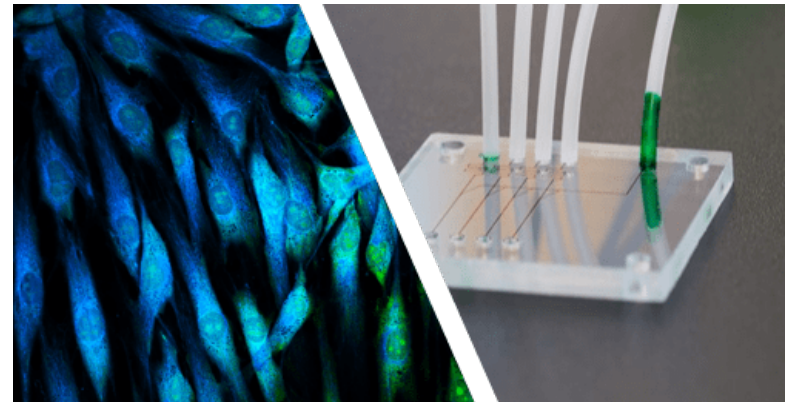
Electrogenic

Passive



Applications:

- Study of neural networks
- Organs on chip
- Drug discovery...





Biological investigation mainly relies on **optical microscopy**

Advantages of direct electrical sensing:

- Label-free (save time and reagents, non intrusive)
- Quantitative
- Integration with microelectronics
- Miniaturization and portability
- Some cells generate electrical signals

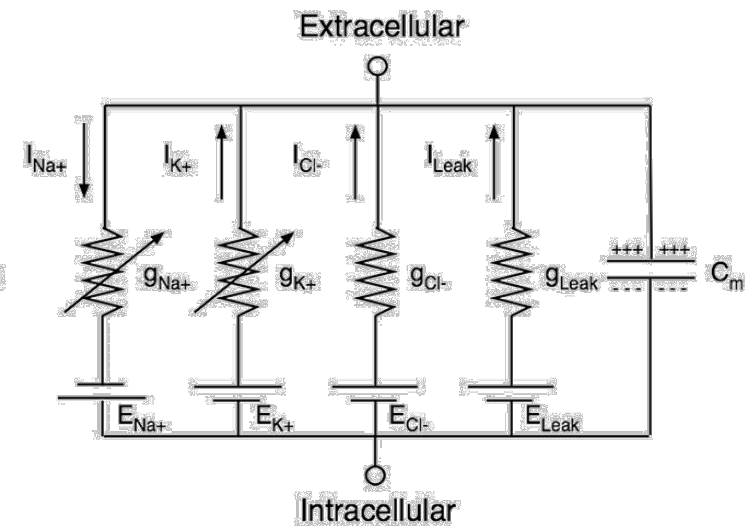
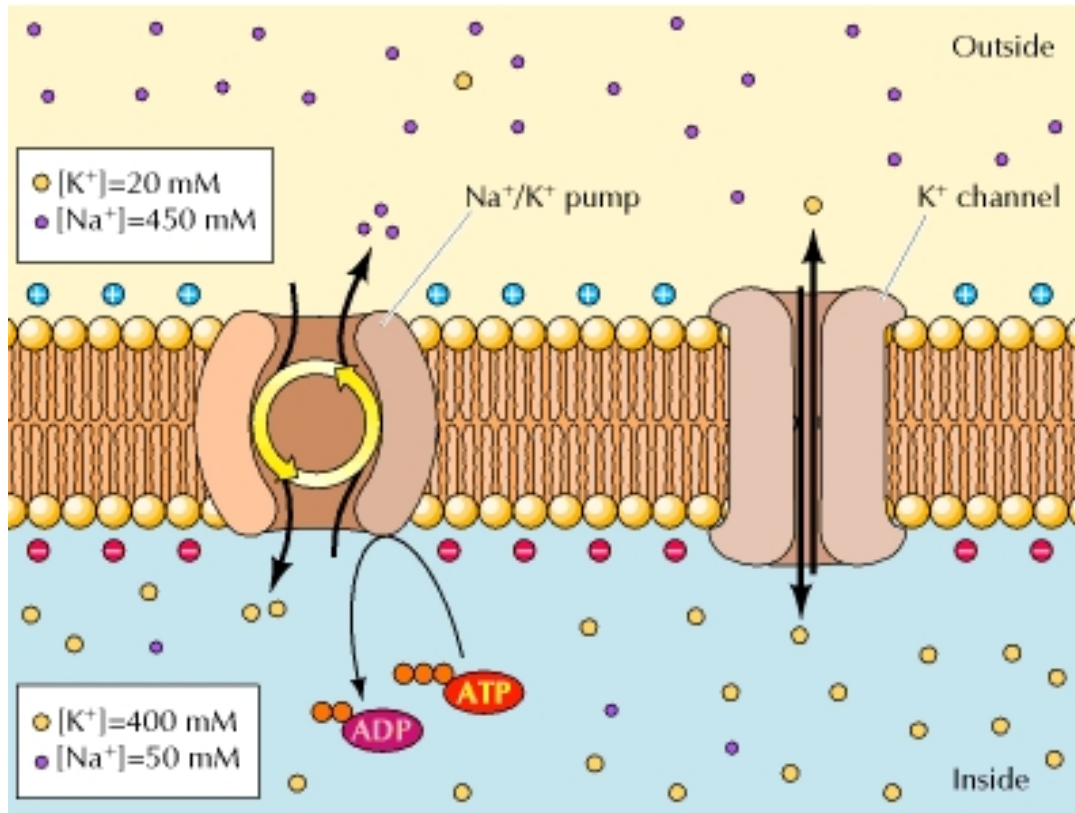
Historical landmarks of electrophysiology:

- 1791 Galvani “*De viribus electricitatis in motu musculari*”
- 1952 **Hodgkin Huxley** action potential model (Nobel 1963)
- 1976 **Patch-Clamp** (by Neher & Sakmann, Nobel 1991)



Electrogenic Cells

Electrical signals within cells are driven by **ionic gradients**:

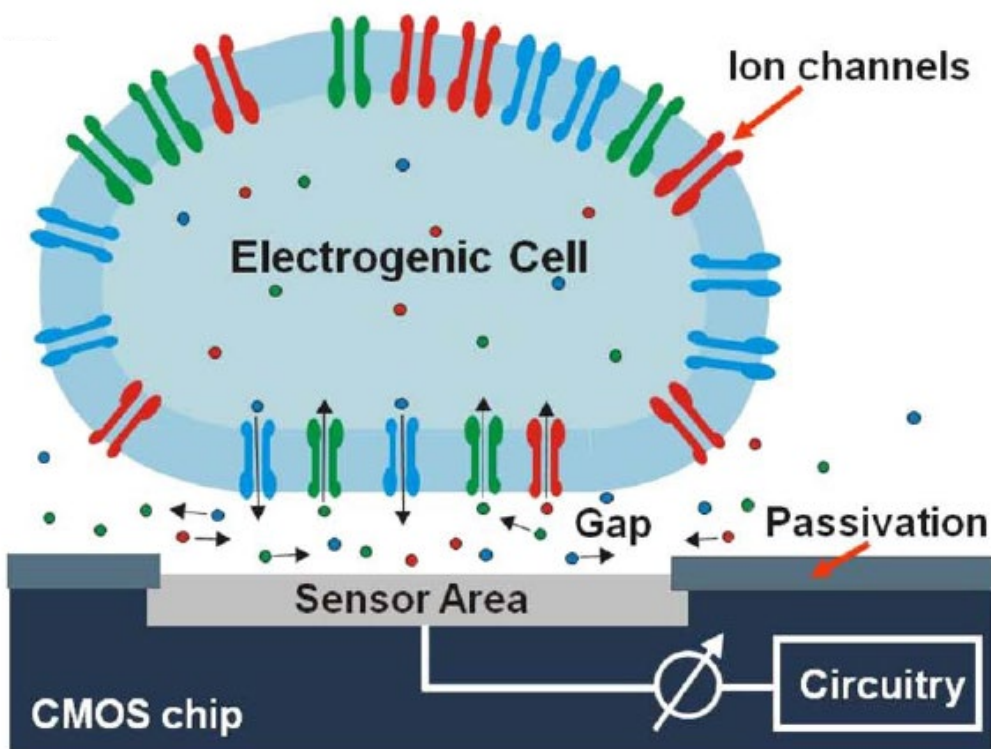


- Pumps (against diffusion, need energy)
- Ion channels (valves)
- Neurons
- Heart, muscles



Complementary Sensing Approaches

Non-invasive external electrical sensing of electrogenic activity:

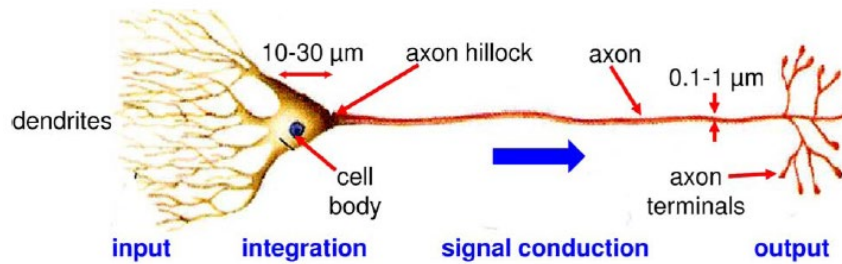


- Voltage
- **Current** (Patch-Clamp)
- **Chemical detection** of released molecules

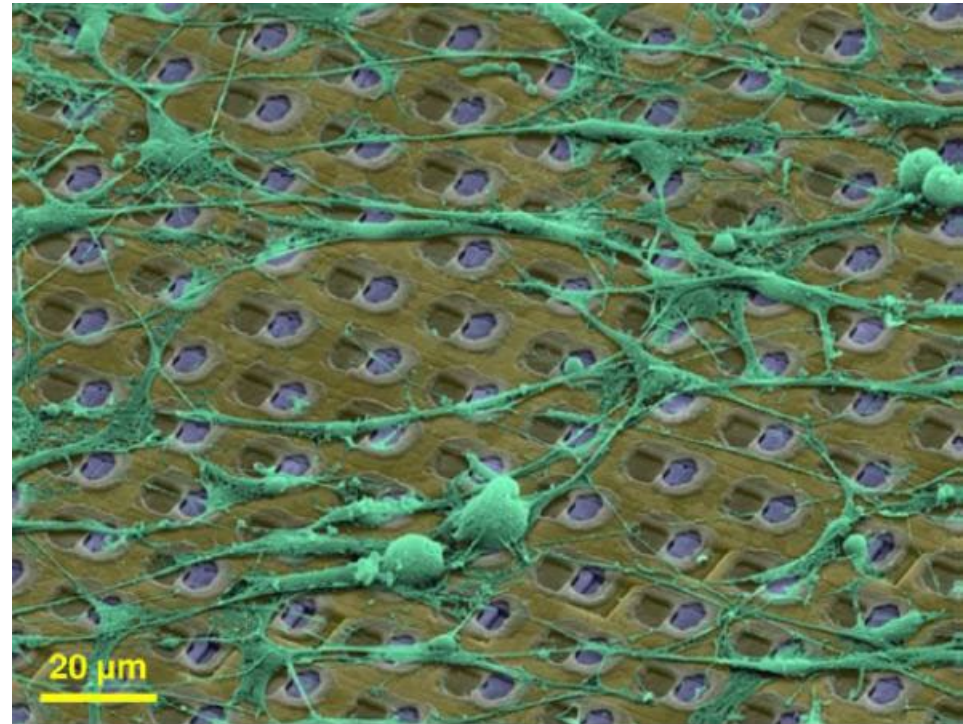
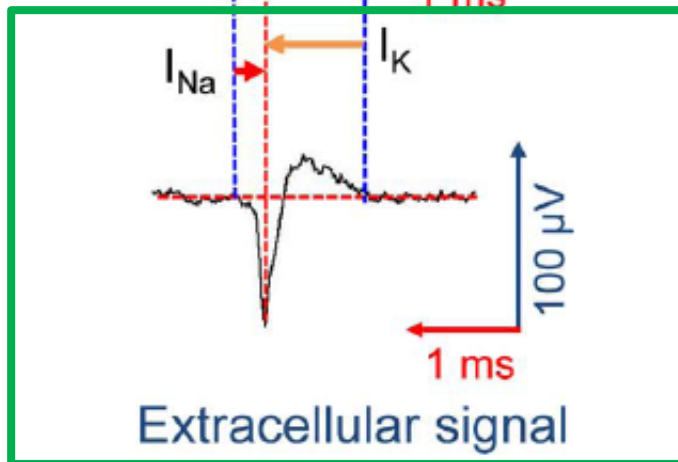
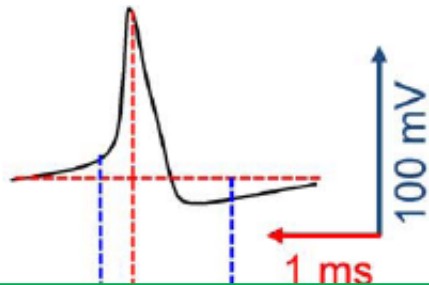
Common trend to miniaturization, parallelization and integration



Voltage Sensing (Neural Networks)



Intracellular signal



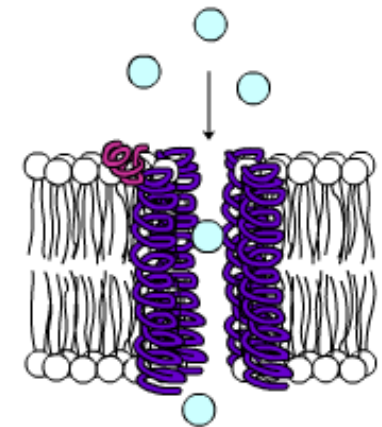
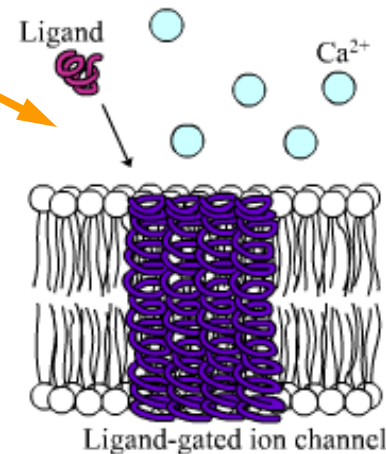
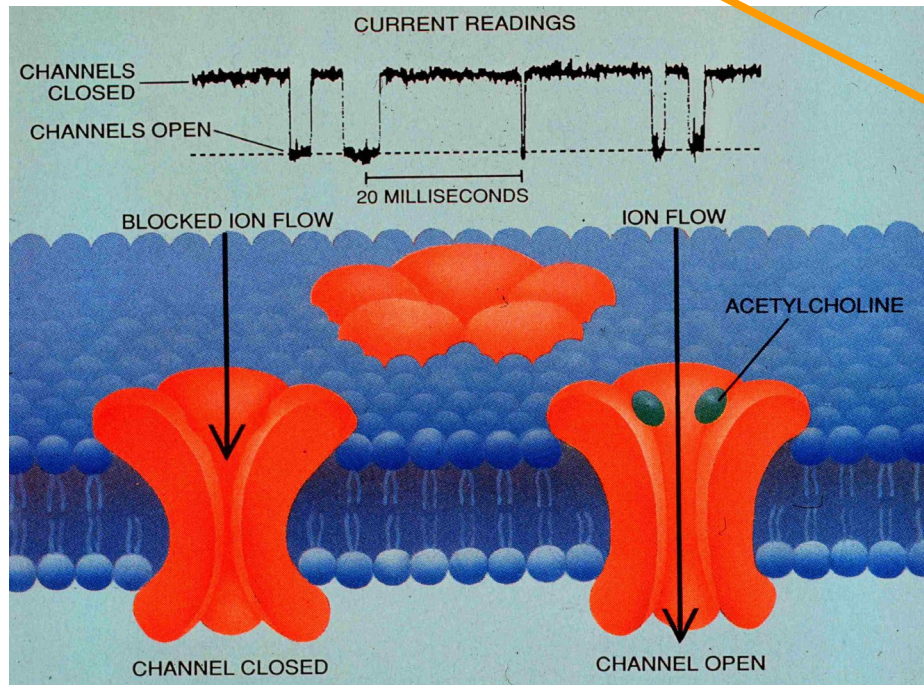
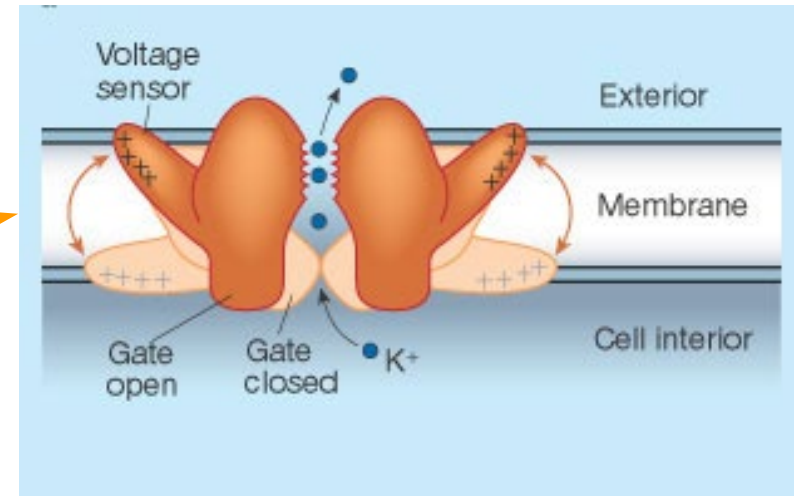
- (Implanted) electrodes are critical
- 10 μV resolution with kHz BW
- CMOS platforms for thousand channels

Life's Transistors: Ion Channels

Ion-specificity

Triggering mechanisms:

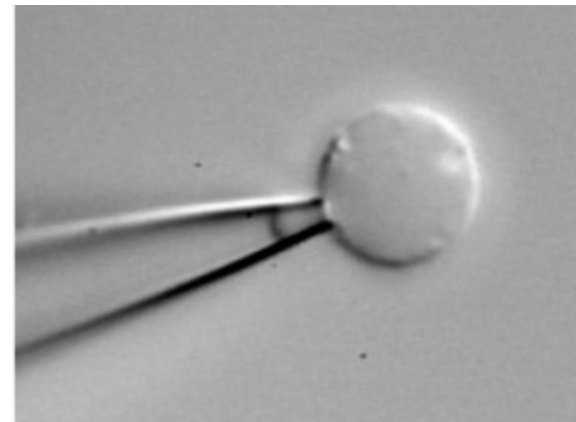
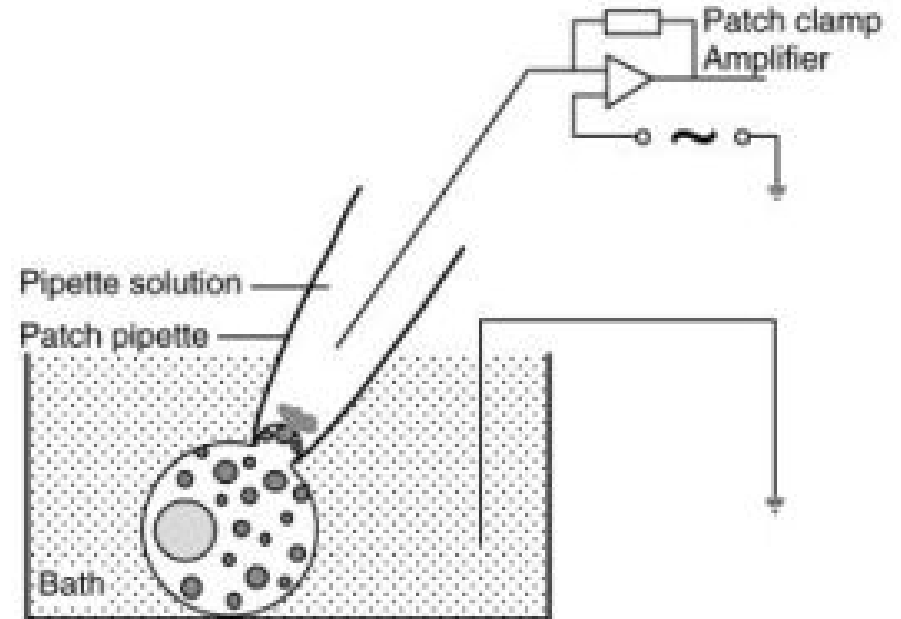
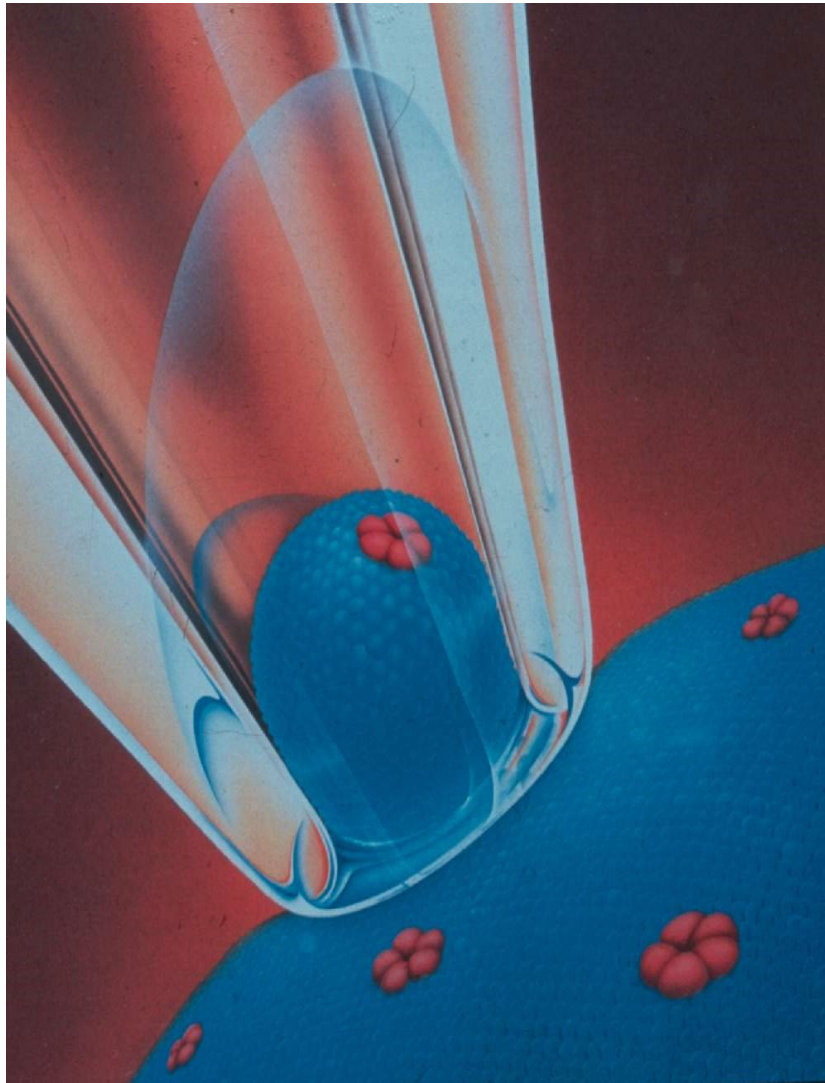
- Voltage-gated
- Ligand-gated



F. Sigworth, *Nature* **423** 2003



The Patch-Clamp



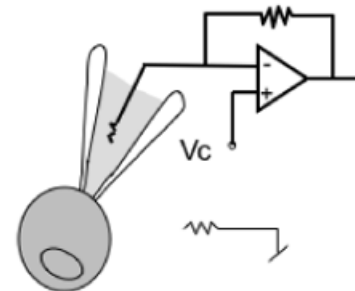
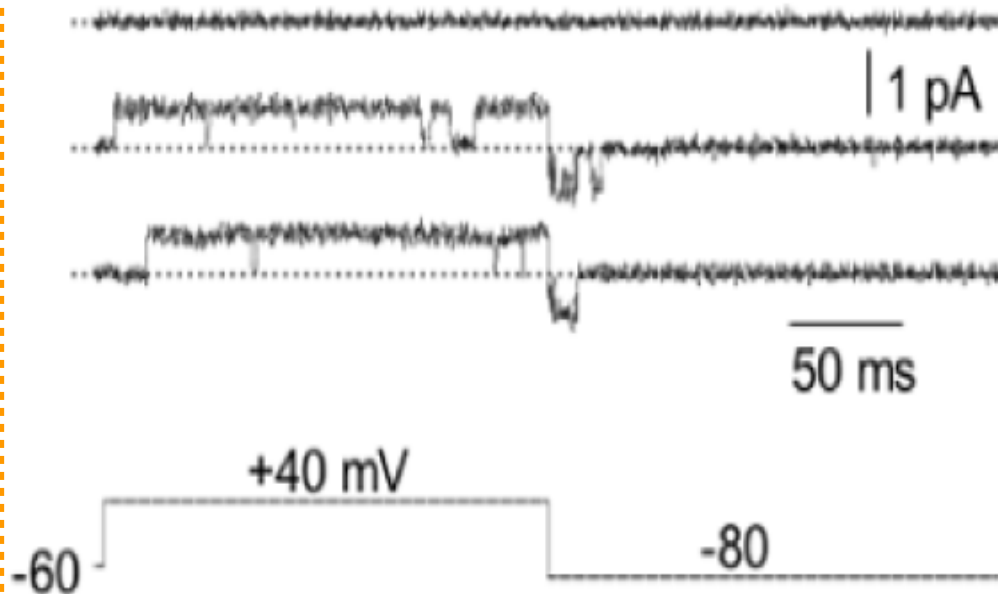


Current Recording

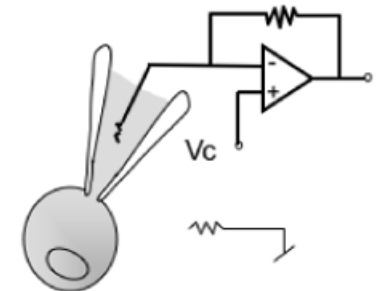
Significant biological interest in improving the **current resolution**:

- Na, K channels 1-10pA
 - Ca, Cl channels 0.1pA
 - Pumps (100e⁻/s) 0.016fA
- } ms timescale

Single channel:



Membrane rupture
whole-cell:



~nA current

F. Sigworth & K. Klemic, *IEEE Trans. Nanobioscience* 4 2005



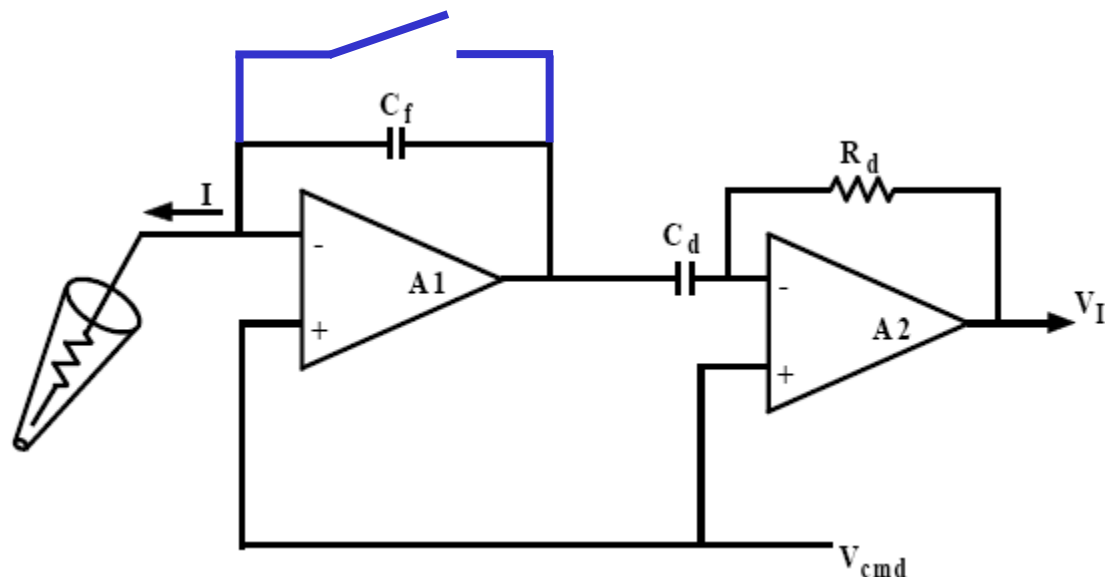
Commercial State-of-the-Art



Axopatch 200B
Axon (Molecular Dev.)

Cooled headstage (-15°C)

Reset

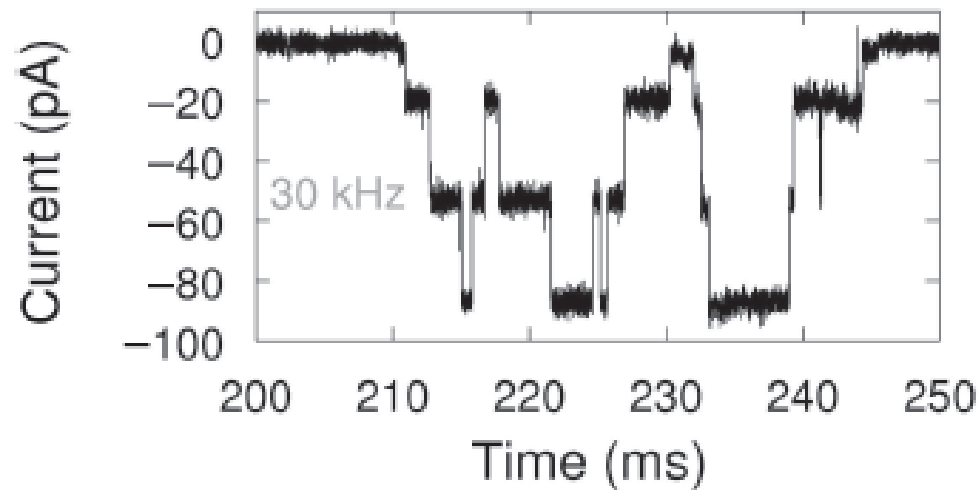
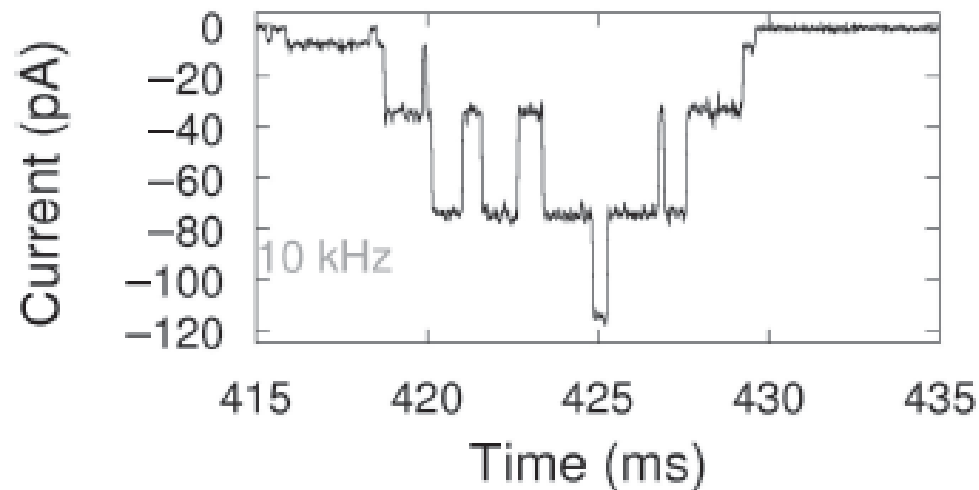


Specs:

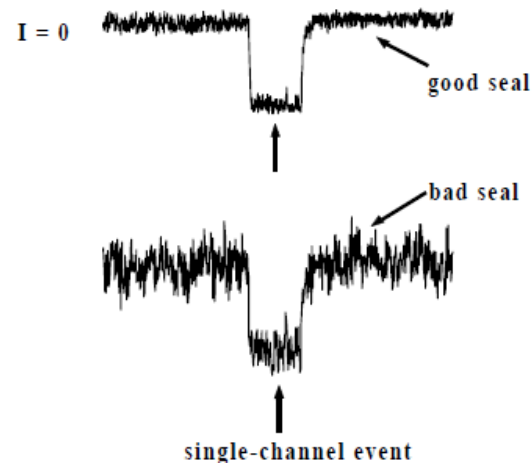
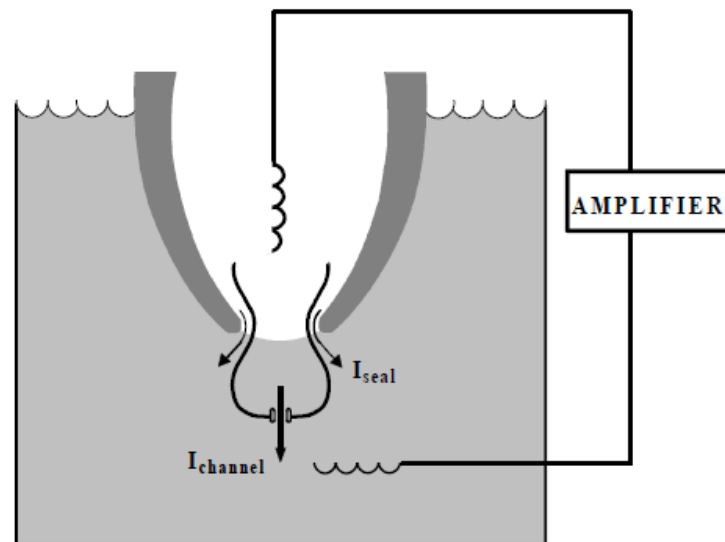
- JFET, $I_{\text{bias}} = 1\text{pA}$
- Gain = 1mV/pA
- BW = 140kHz
- 60fA_{rms} over 5kHz
- $C_f = 1\text{pF}$
- $T_{\text{reset}} = 50\mu\text{s}$



Achievable Performance

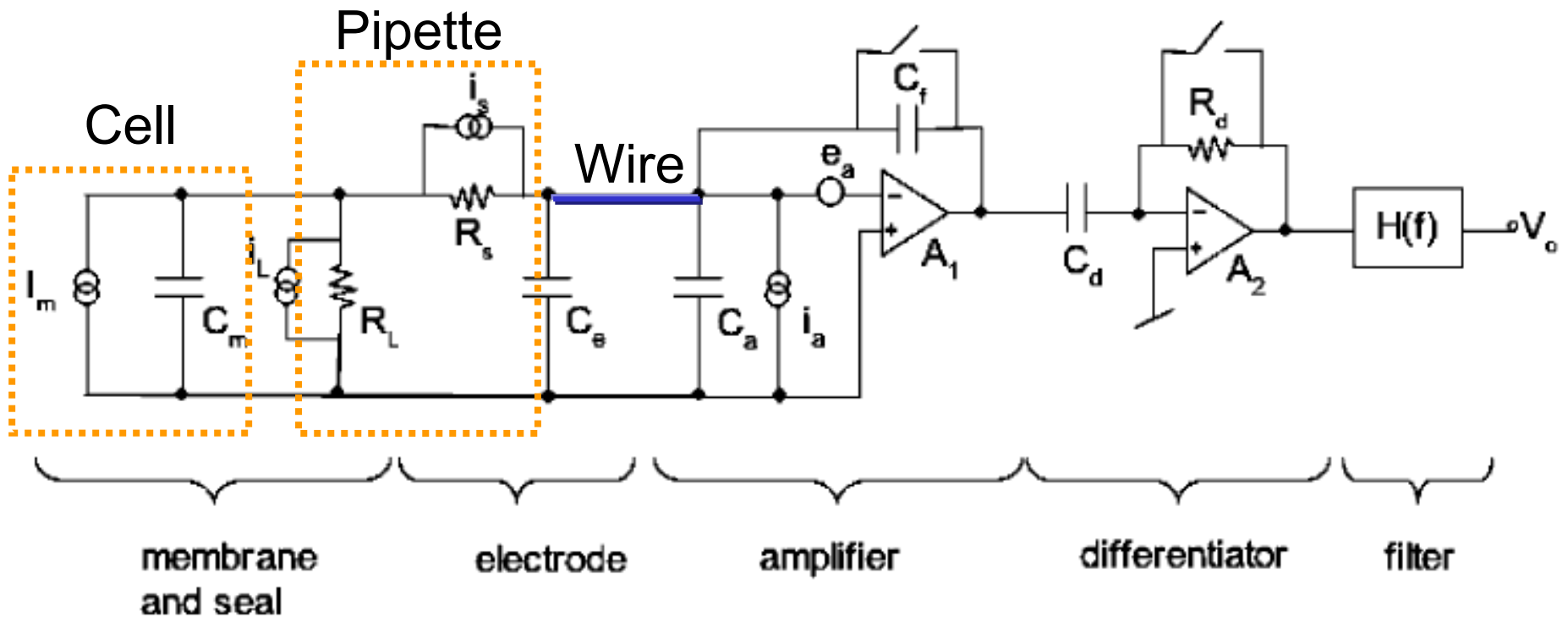


Careful sealing



How to Improve the Performance

The dominating noise terms are due to the **setup**:



- $C_m = \sim 0.01 \text{ pF}/\mu\text{m}^2$
- $R_L = \sim \text{G}\Omega$ (seal)
- $R_S = \sim \text{M}\Omega$ (solution)
- $C_e = 1\text{-}10 \text{ pF}$
- $C_a = 1\text{-}10 \text{ pF}$

F. Sigworth & K. Klemic, *IEEE Trans. Nanobioscience* 4 2005



Improving the Setup

Pipette improving:

- Quartz
- Polymeric coating

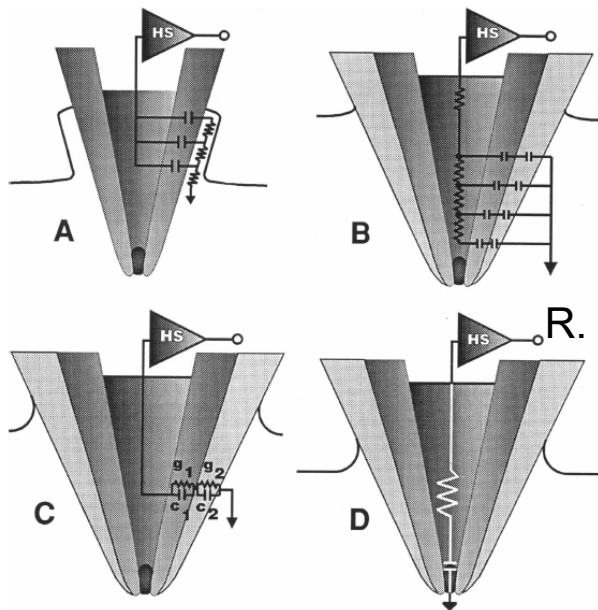


avoid pipette

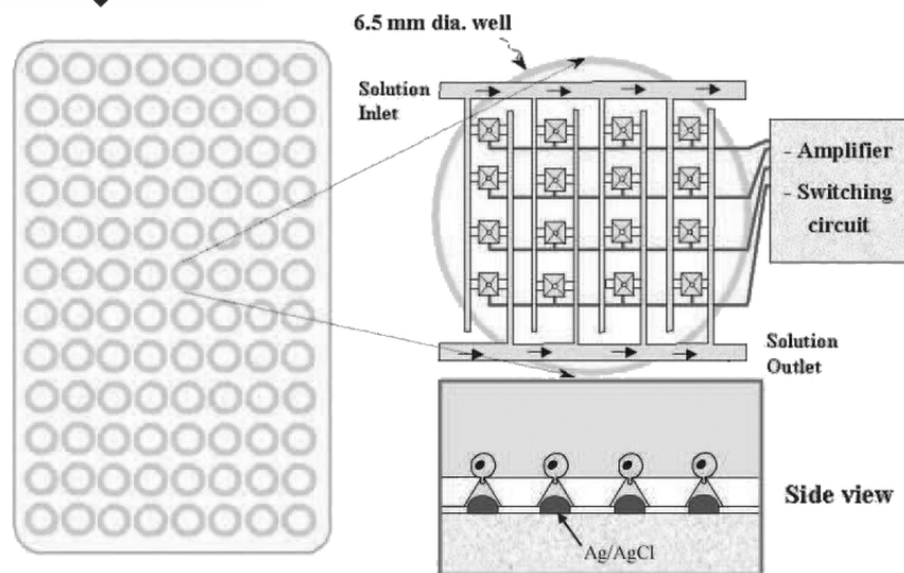
Planar setup:

- Reduce stray
- Automatic
- Parallel

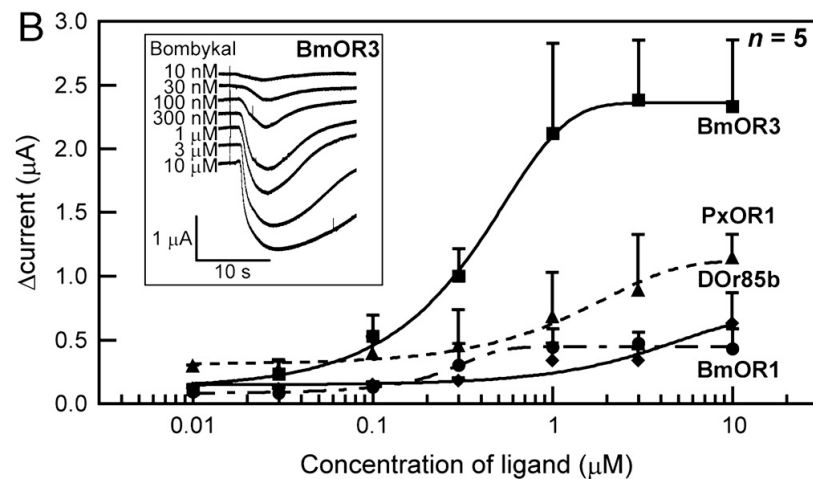
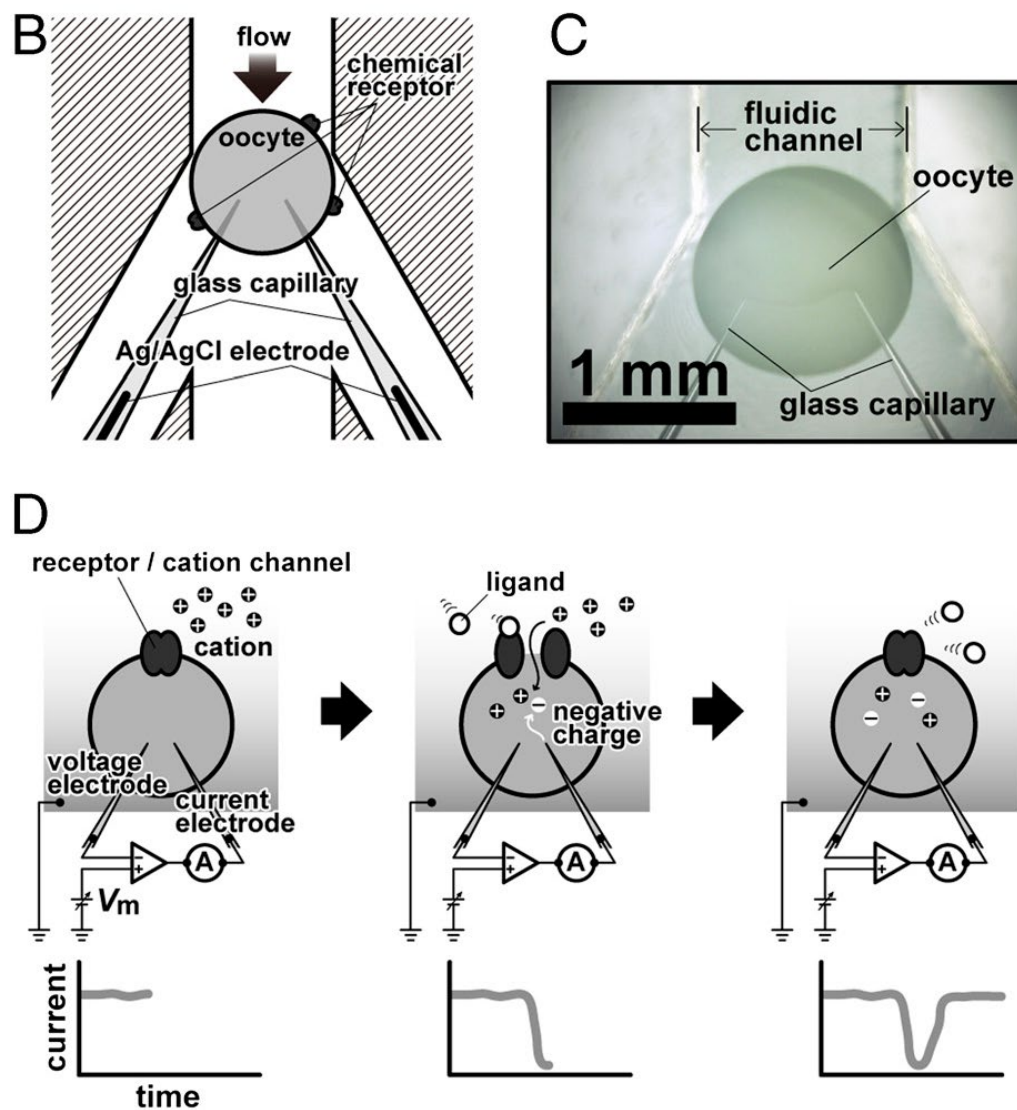
K. Klemic *et al.*, *Biosens. Bioelectron.* **17** 2002



R. Levis & J. Rae, *Biophys. J.* **65** 1993



Whole Live Cell Odorant Sensor

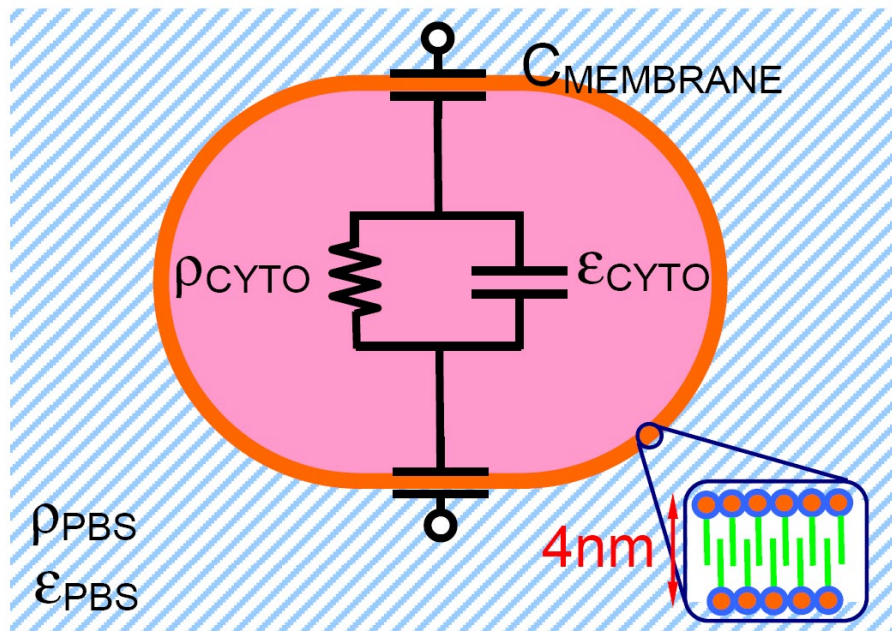


N. Misawa *et al.*, PNAS **107** (2010)

Passive Electrical Properties

As the passive electrical properties are probed, **any kind of cell** can be detected

Small signal equivalent: the single shell model

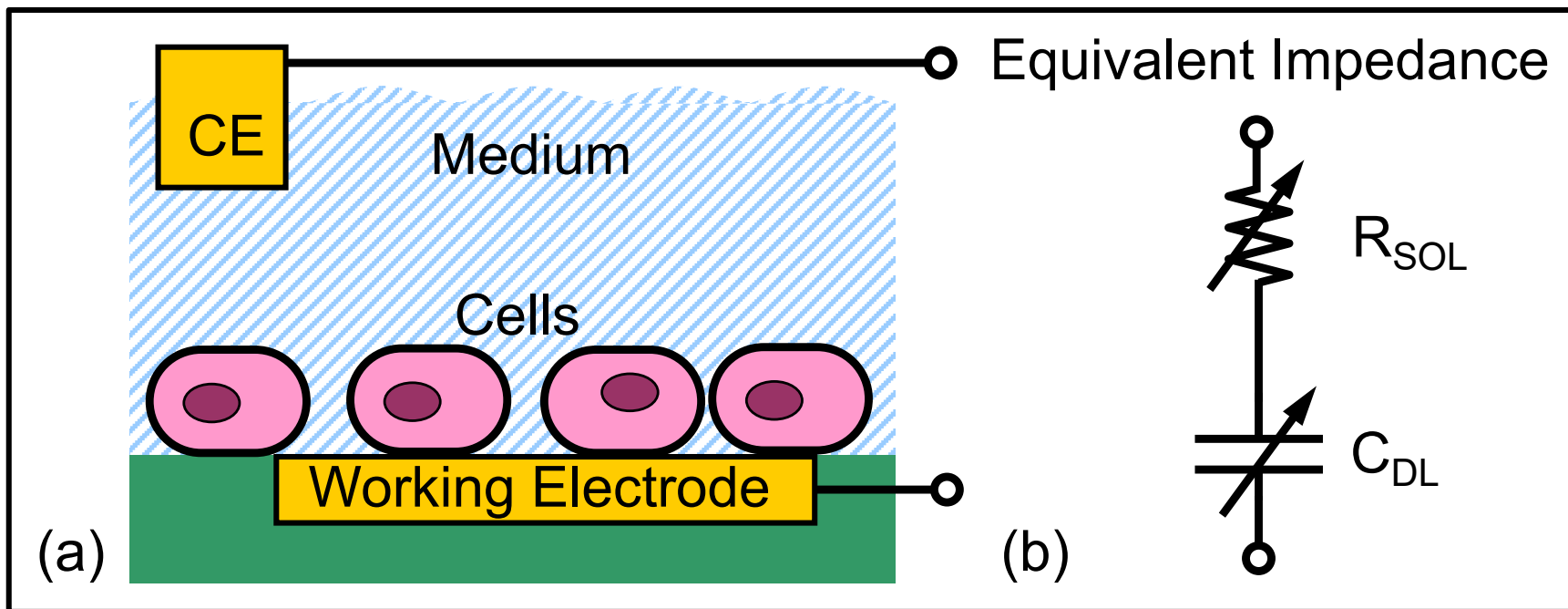


- $\rho_{PBS} = 66\Omega\text{cm}$
- $\epsilon_{PBS} = 78$
- $C_{MEM} = 0.01\text{pF}/\mu\text{m}^2$
- $\rho_{CYTO} \sim \rho_{PBS}$
- $\epsilon_{CYTO} = 60$

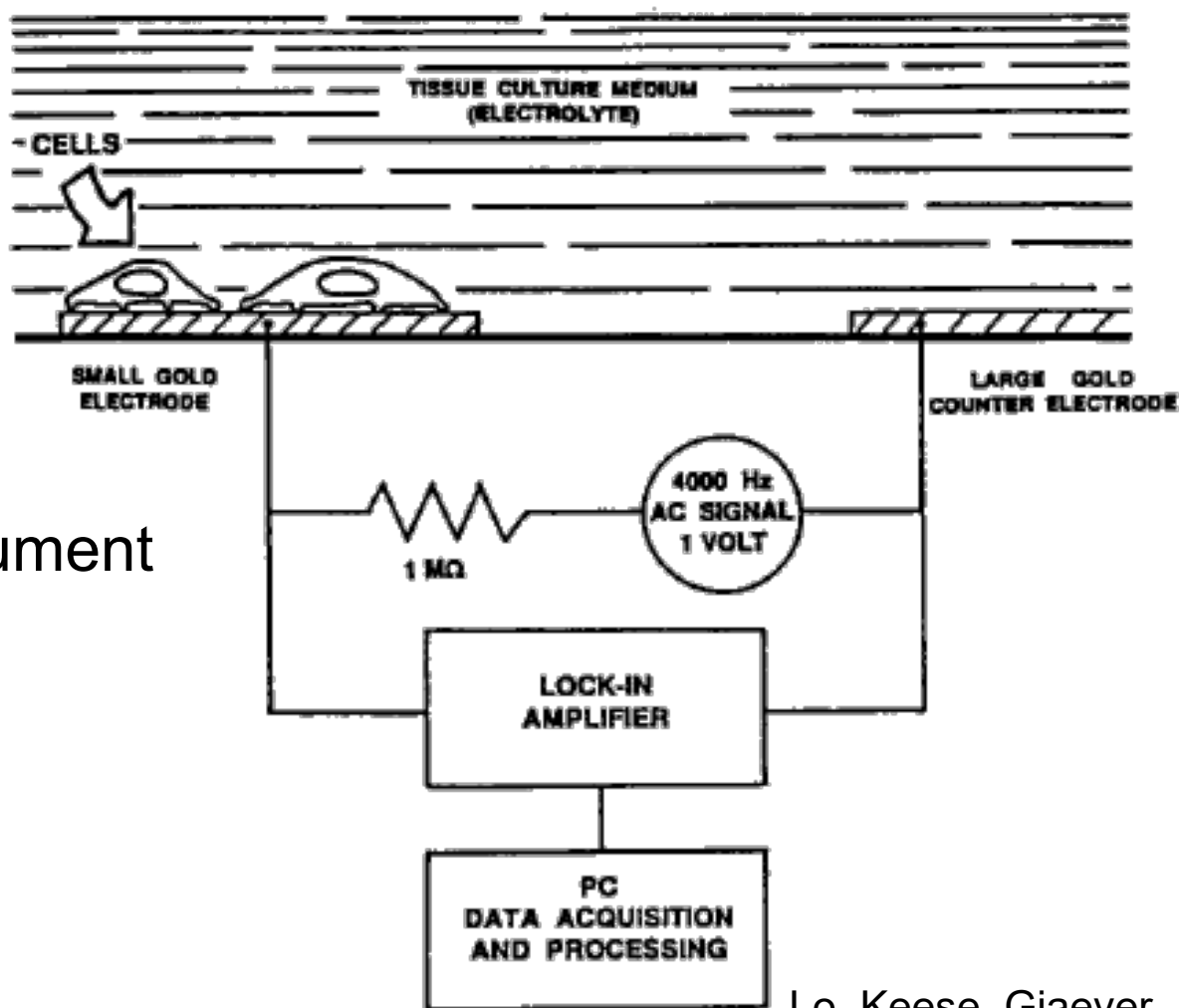
- Low frequency (below $\sim 100\text{kHz}$): insulating sphere
- Dead cells (*lysis*, broken membrane) are hardly discernable



Cells on the Electrochemical Interface



Electric Cell–Substrate Impedance Sensing



The instrument

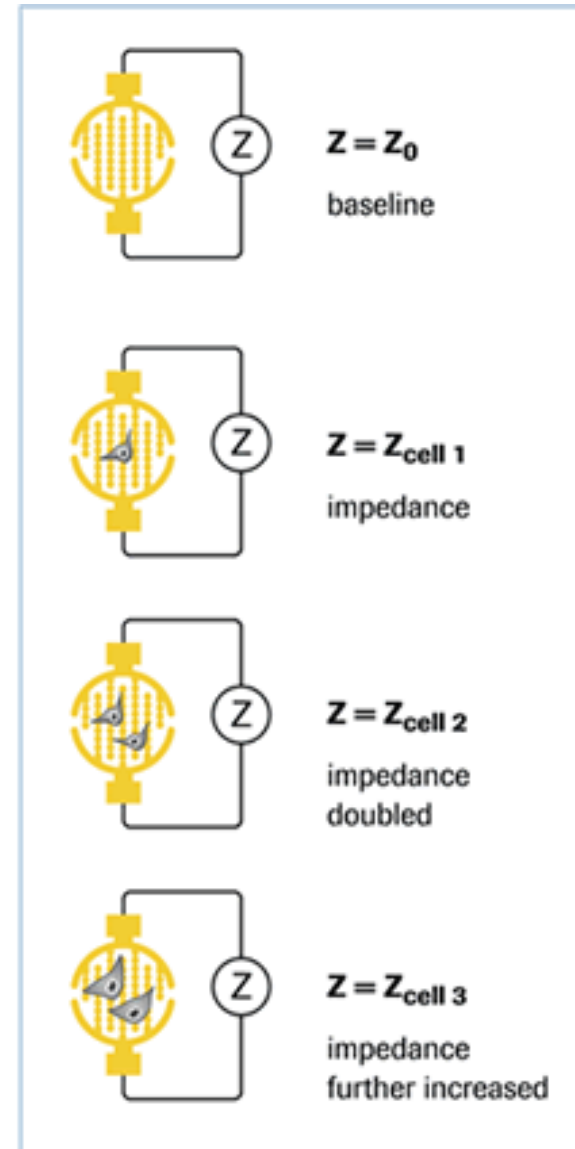
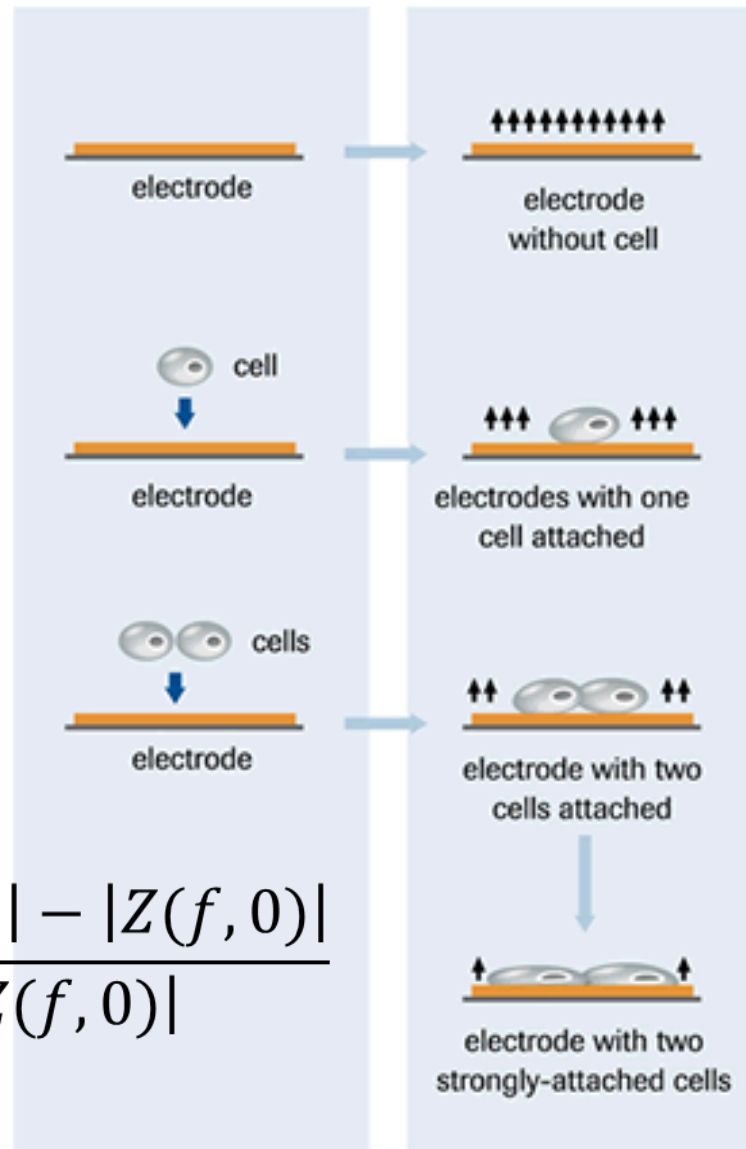
Lo, Keese, Giaever, *Biophys J.* **69** 1995

The Idea: Adding Stuff on Top

Tracking
normalized
impedance
variations

Cell Index

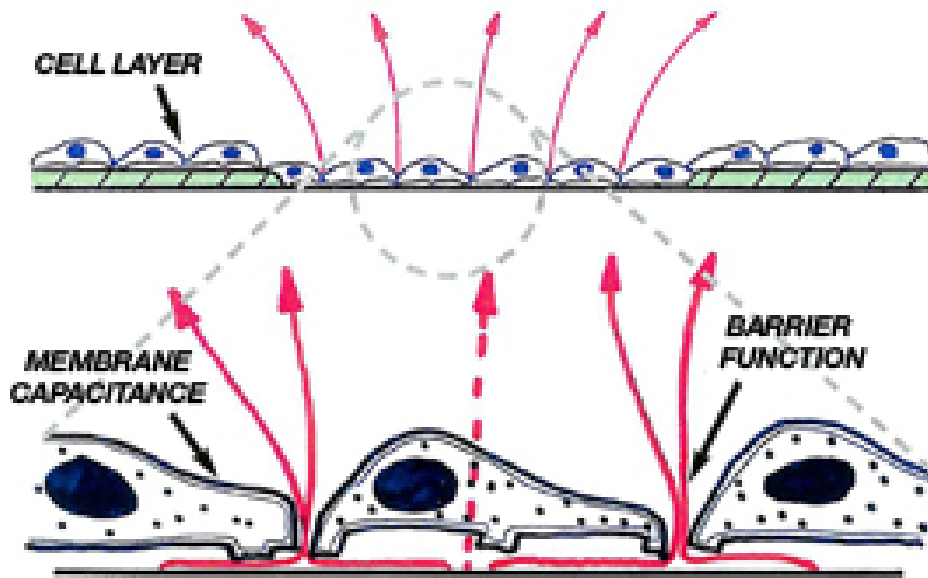
$$CI(f, t) = \frac{|Z(f, t)| - |Z(f, 0)|}{|Z(f, 0)|}$$



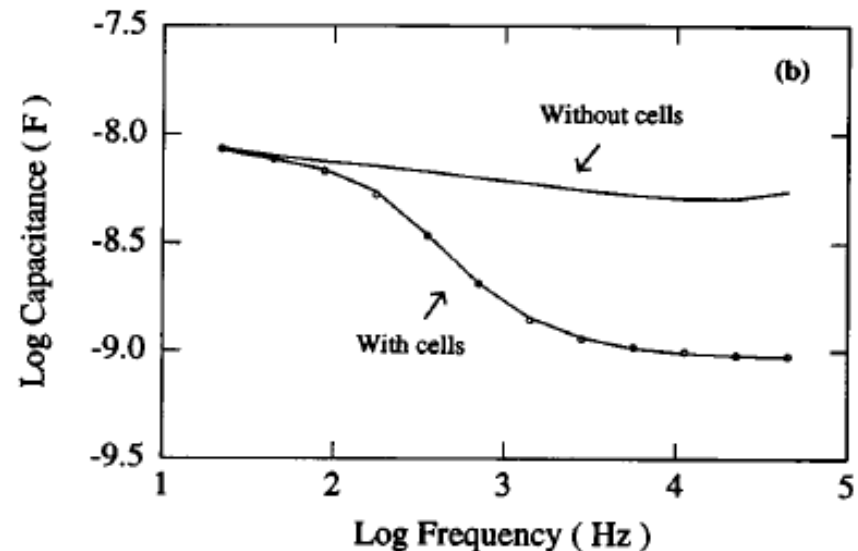
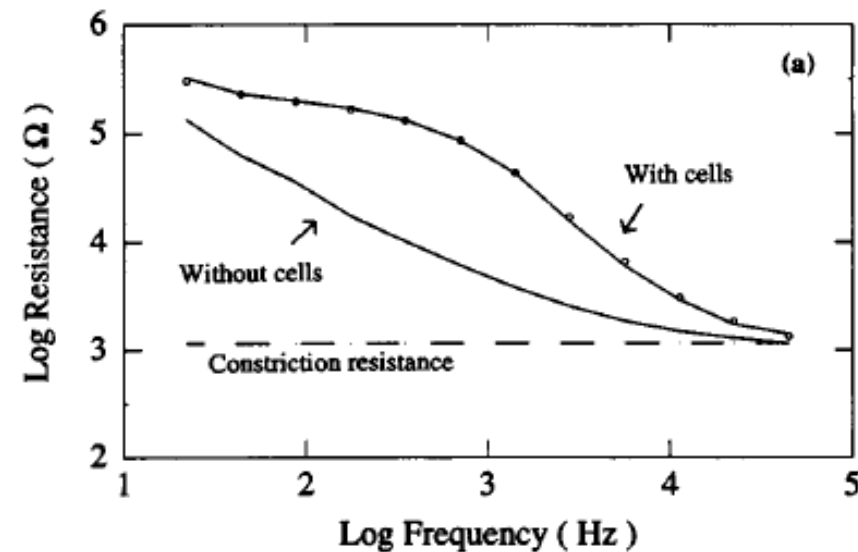


ECIS: Change of Interface Impedance

The model:



- Resistance increases
- Capacitance decreases





Which Processes Can Be Tracked?

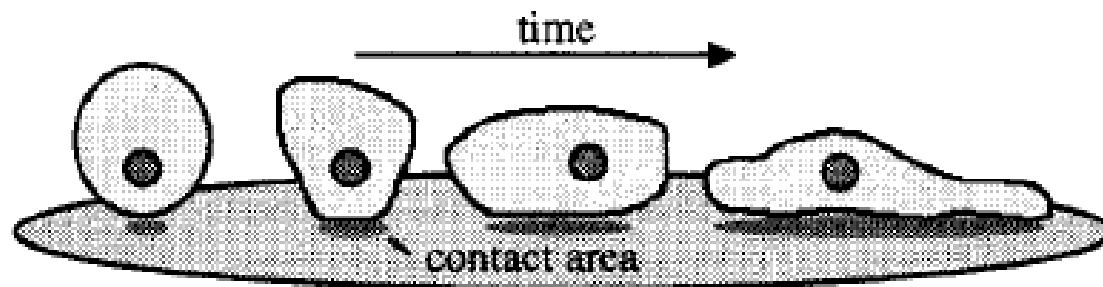
Different mechanism are taking place:

- Sedimentation
- Adhesion and spreading
- Growth and reproduction
- Response to chemo-mechanical stimuli
- Confluence
- Death



The contact area changes over time

Average population information





Effect of Adhesion

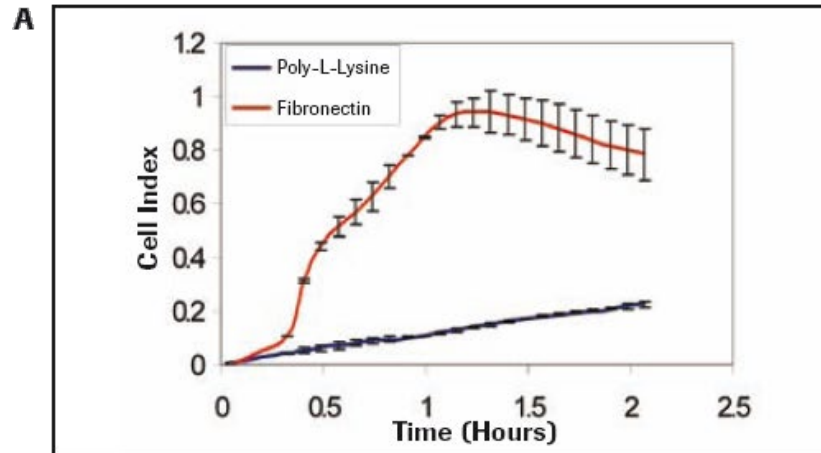
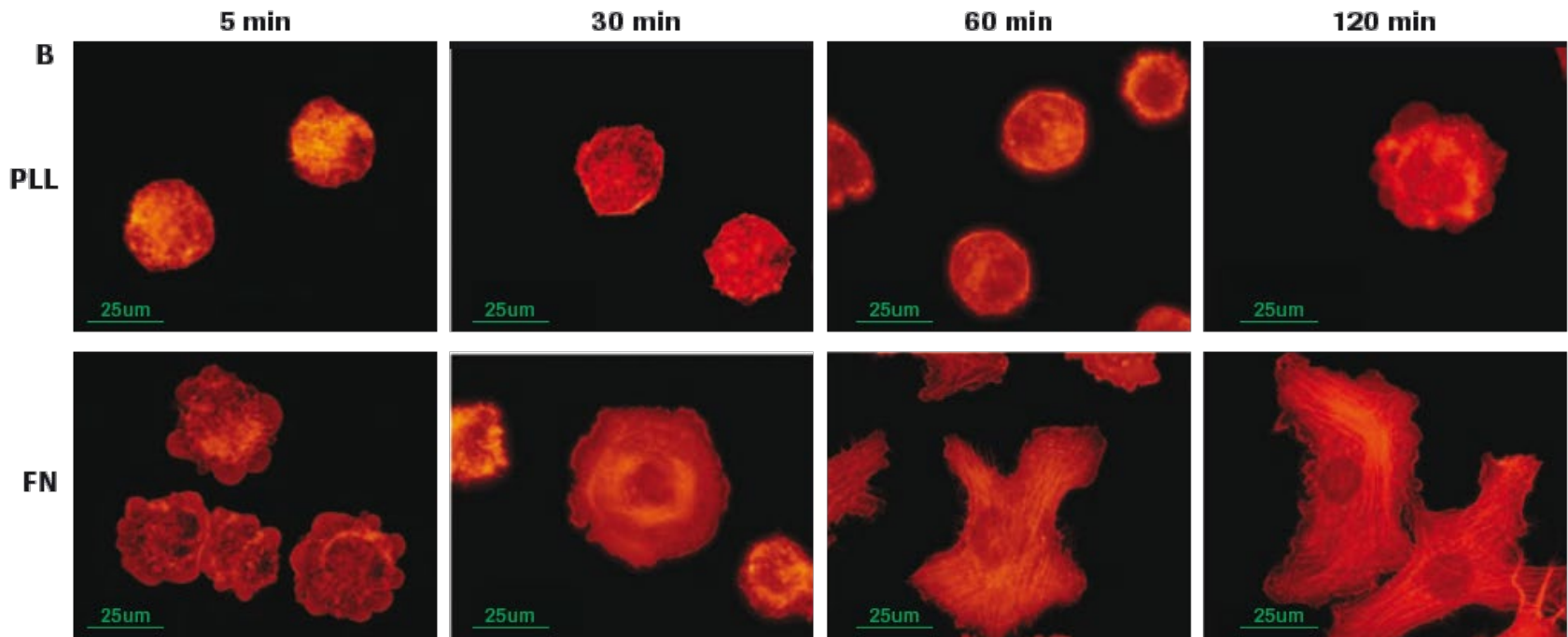


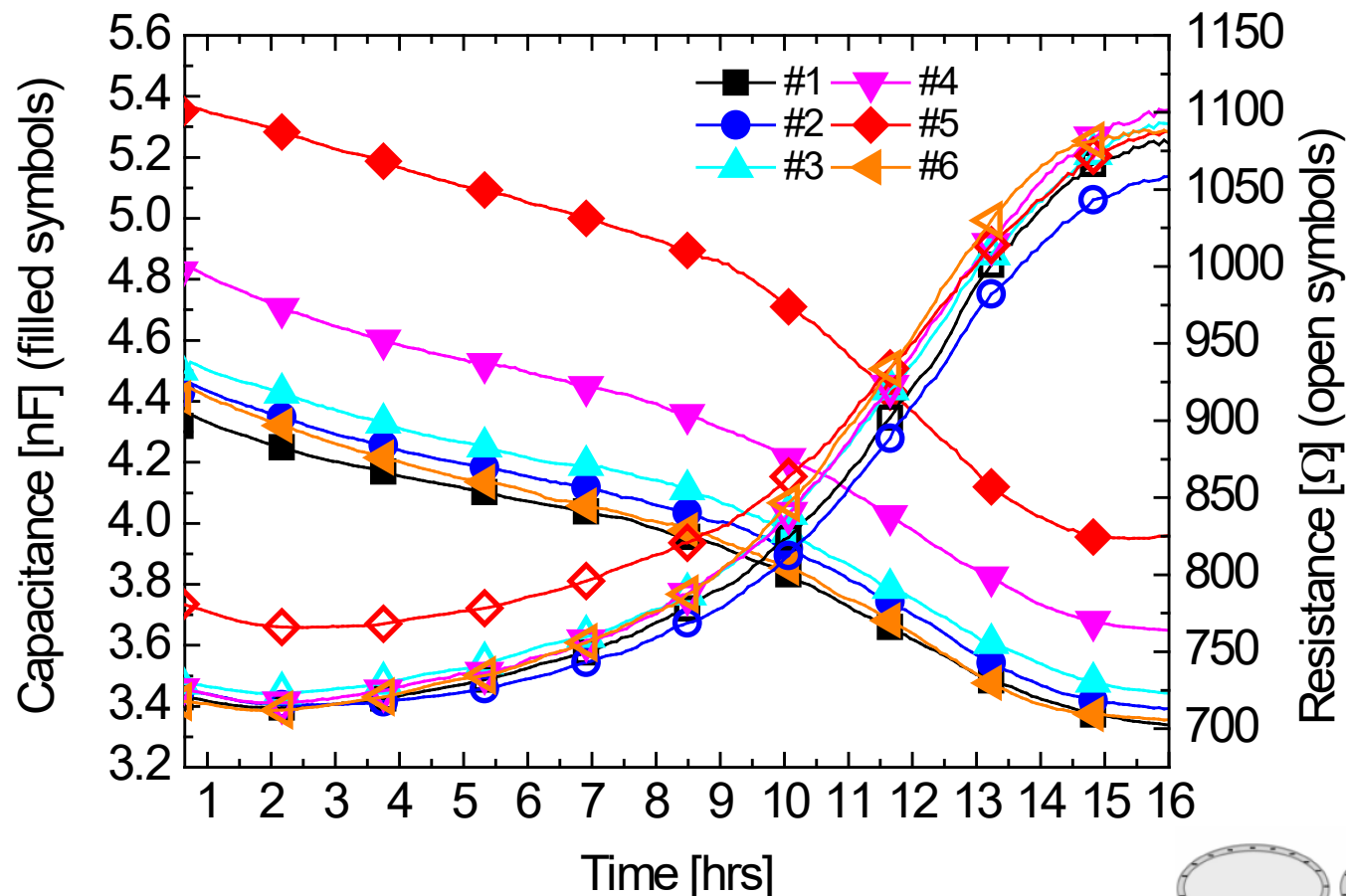
Figure 1. (A) Dynamic monitoring of cell attachment and spreading on PLL and FN-coated surfaces. (B) The Cell Index correlates with the extent of cell attachment and spreading observed using conventional phalloidin staining of the actin cytoskeleton and immunofluorescence microscopy.





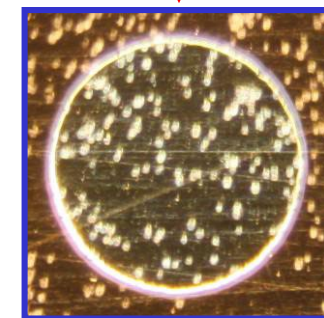
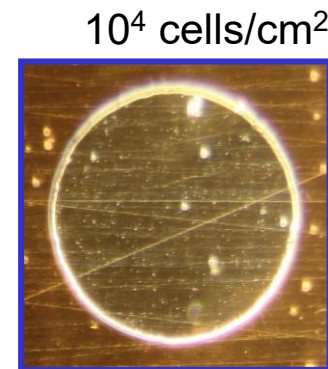
Tracking Yeast Proliferation

Measured: $\Delta \text{Im} = -25\%$ $\Delta \text{Re} = +45\%$

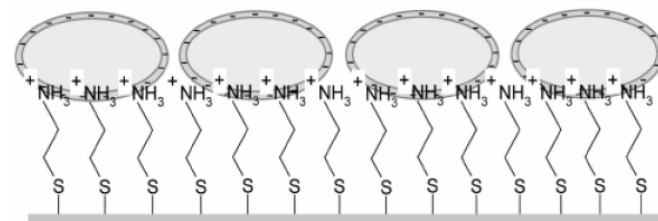


@ 4kHz

In agreement w/ Heiskanen, *Langmuir* 2008

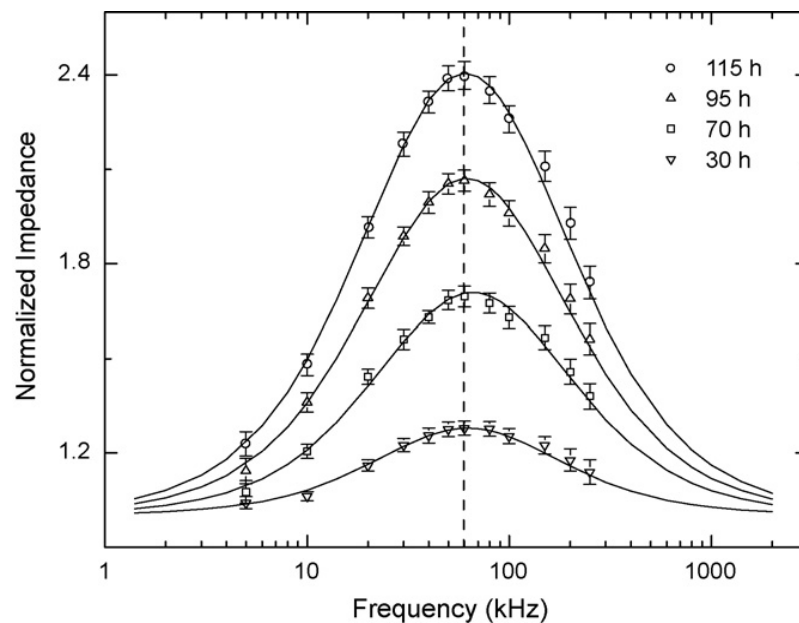
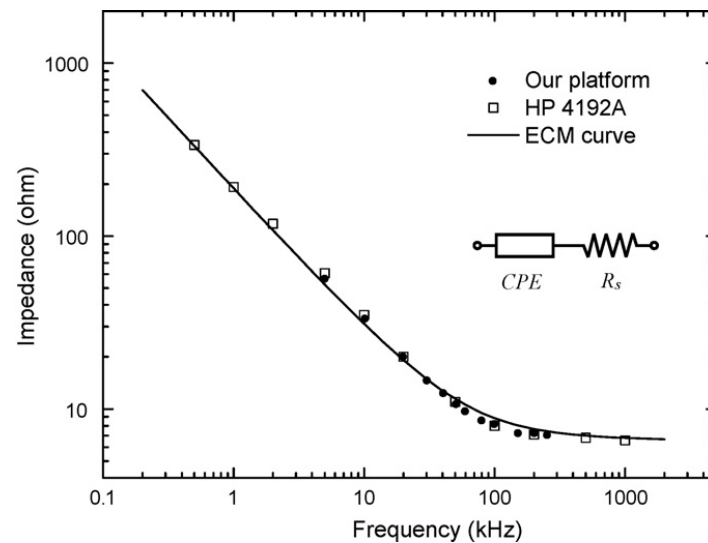
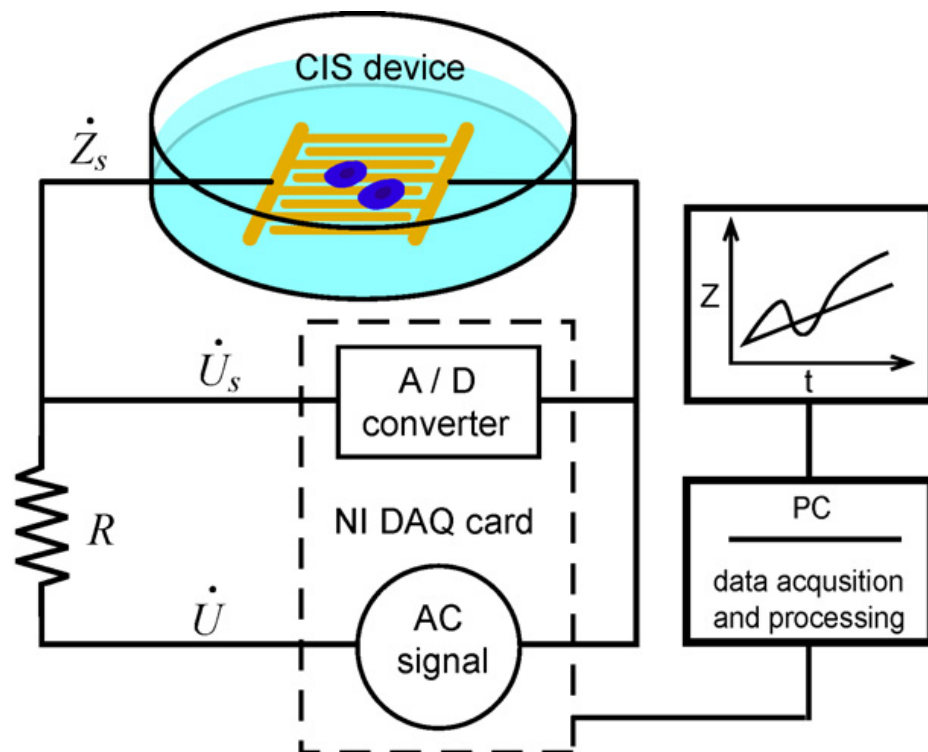


250 μm Gold disk + Cys





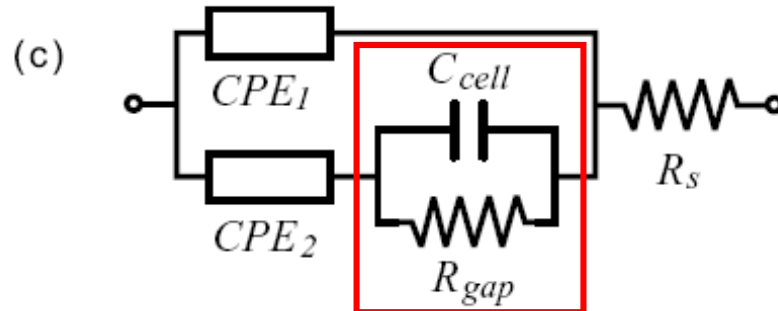
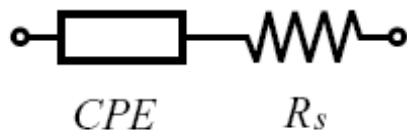
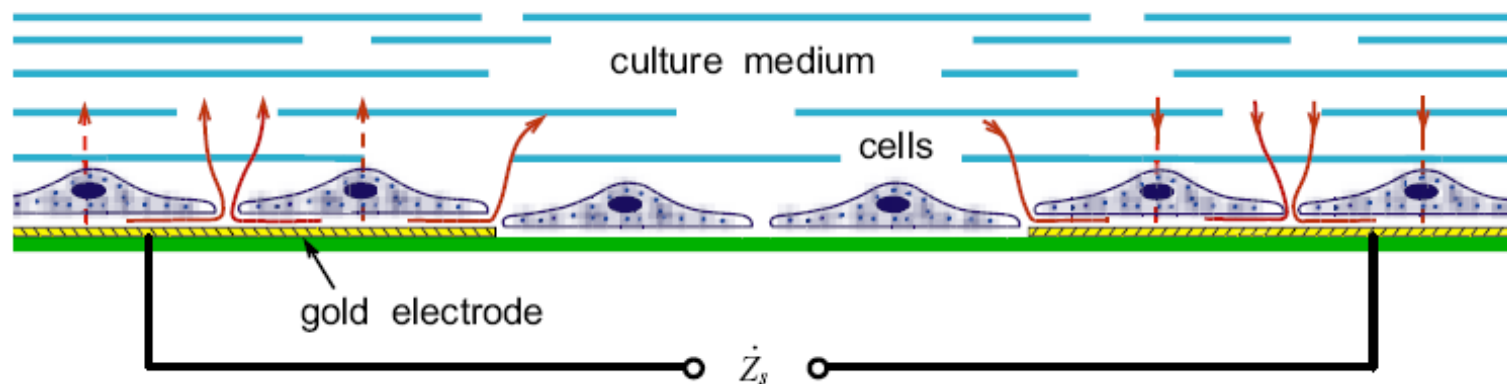
Another Example



Wang et al., *Biosensors Bioelectron.* **25** 2010



Equivalent Model

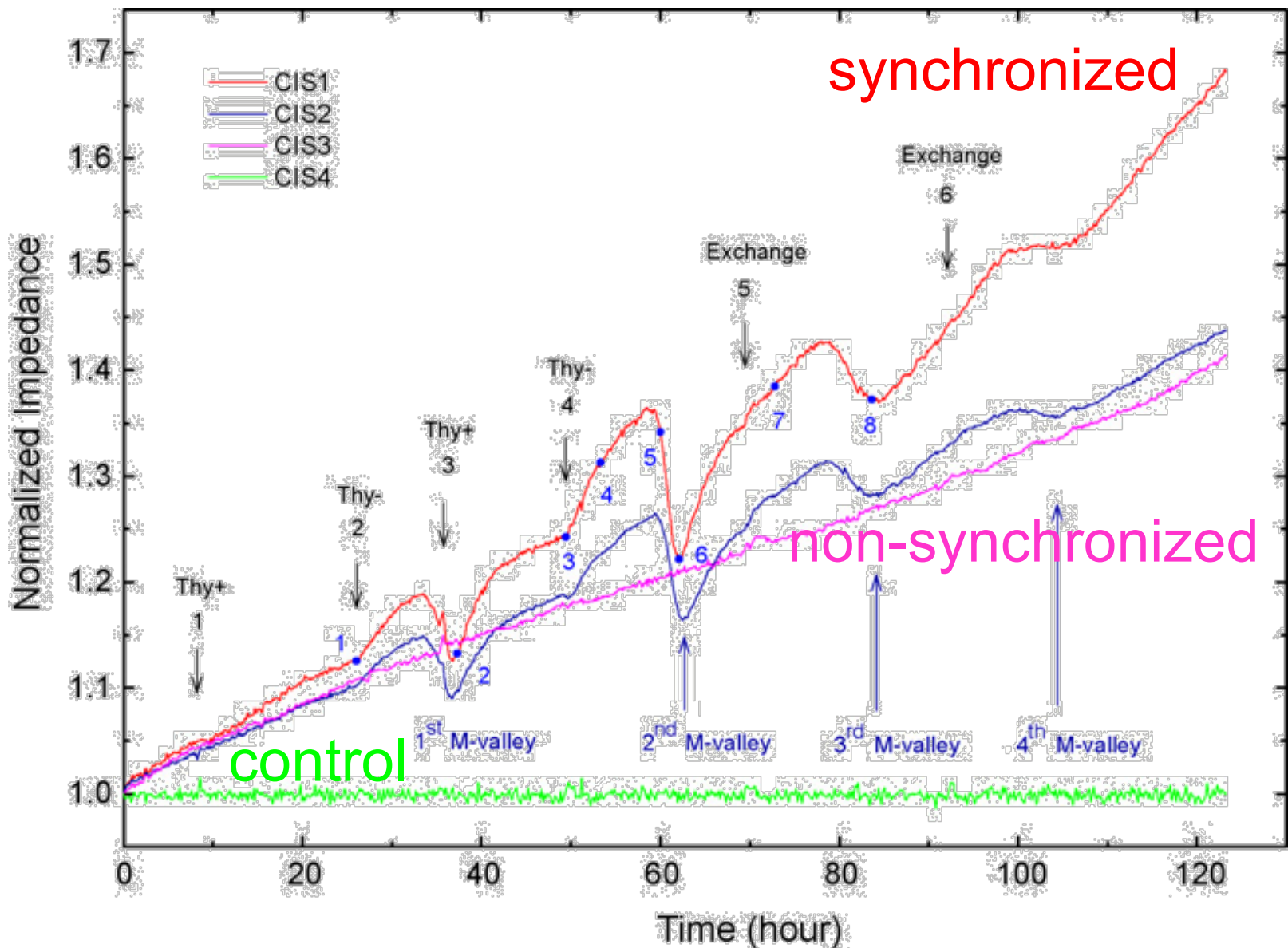


r is the ratio of the total surface over the covered surface by adherent cells

$$Z_{CPE1} = Z_{CPE} / (1 - r) \quad \text{uncovered}$$

$$Z_{CPE2} = Z_{CPE} / r \quad \text{covered}$$

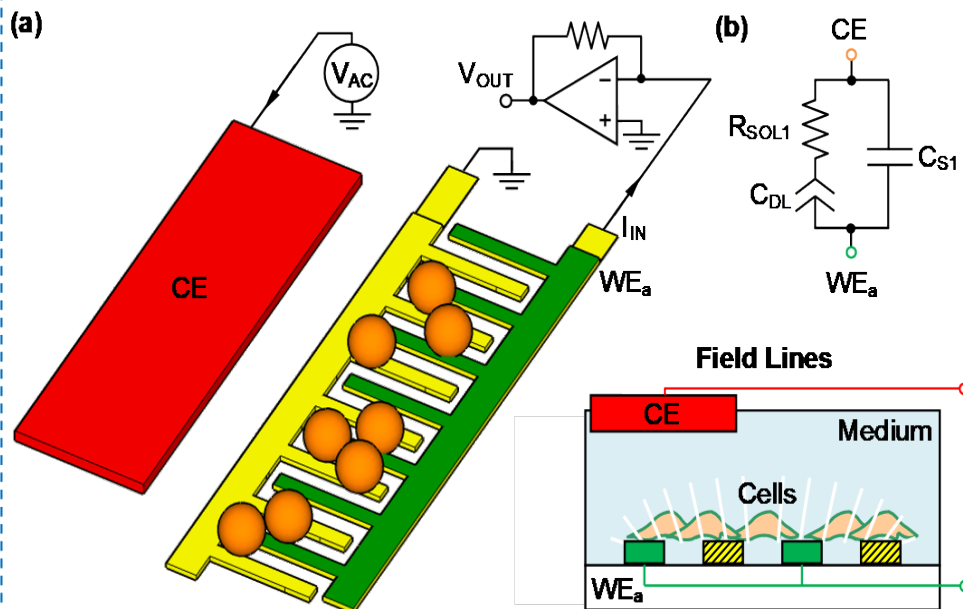
Evidence of Synchronization



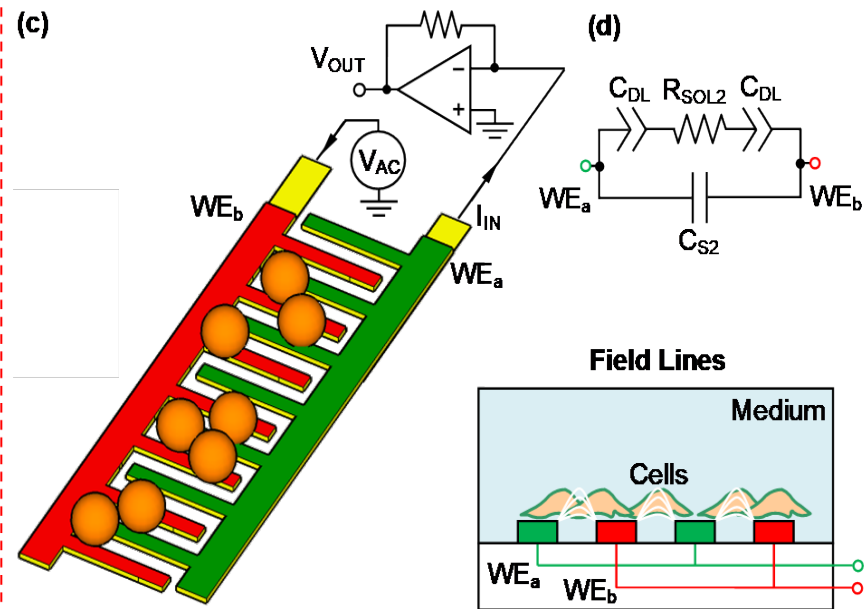


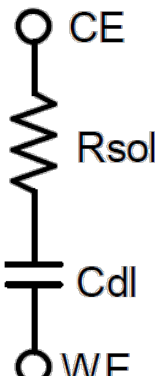
Comparison of Sensing Geometry

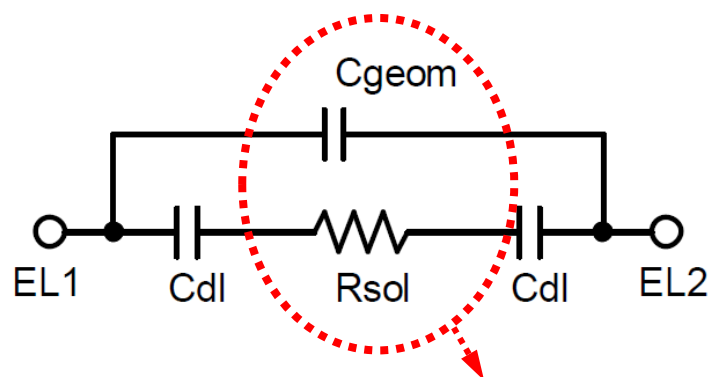
Vertical Sensing Configuration and Equivalent Impedance Model



Coplanar Sensing Configuration and Equivalent Impedance Model



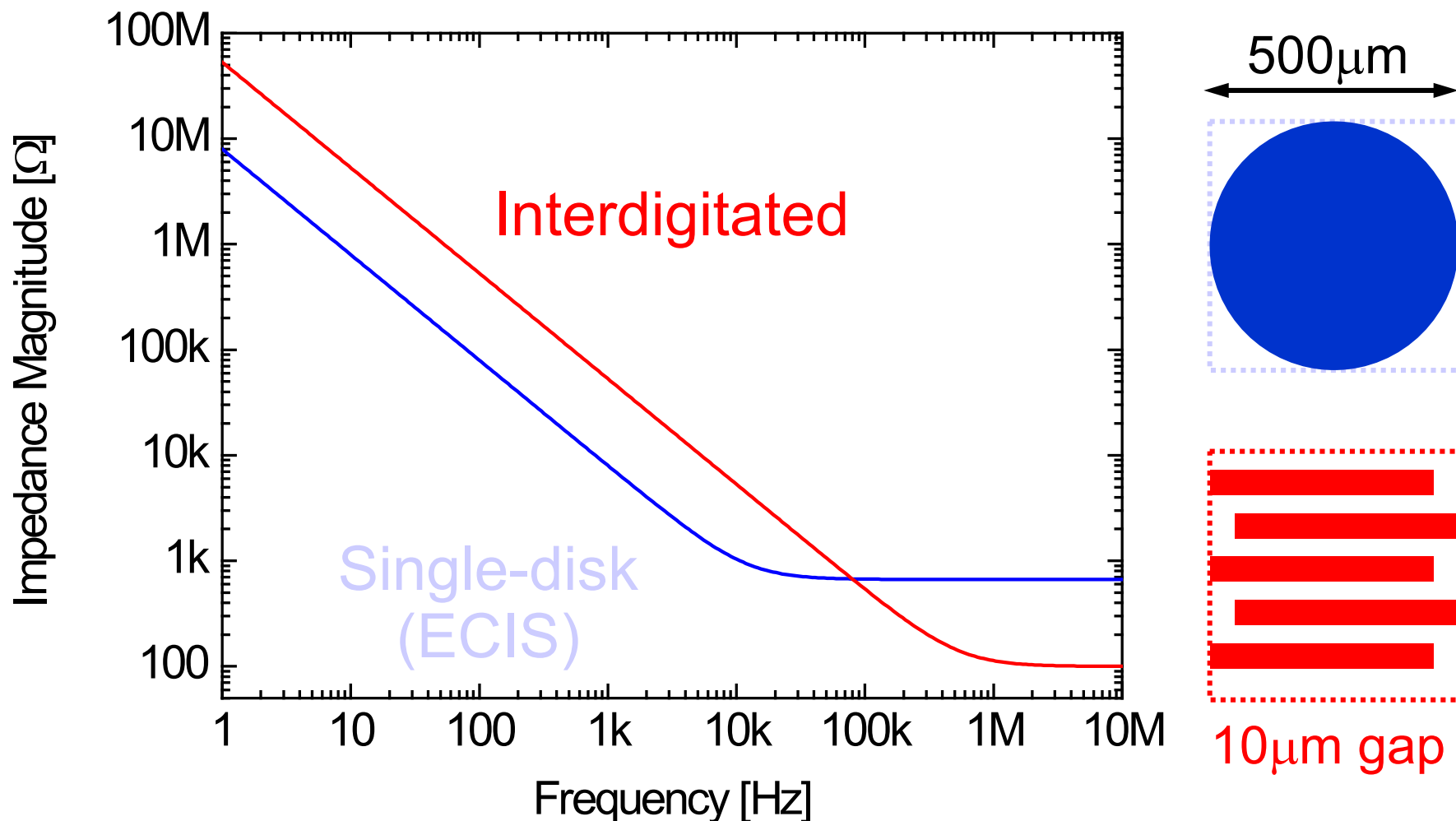
$$R_{sol} = \frac{\rho}{4r}$$
$$C_{dl} = C_0 \cdot \pi \cdot r^2$$




Conformal Mapping



Simulation

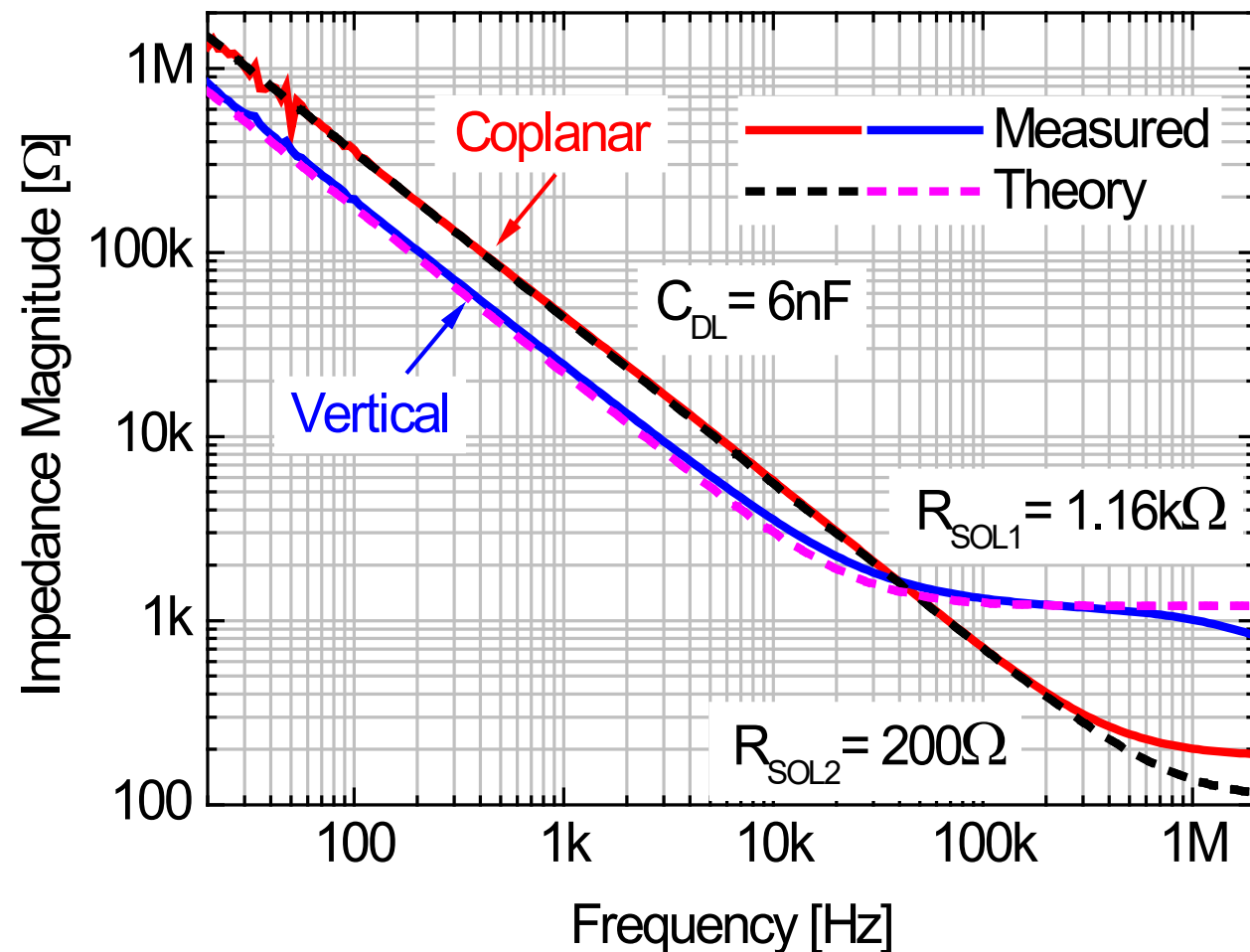


The best operating frequency seems to be the corner



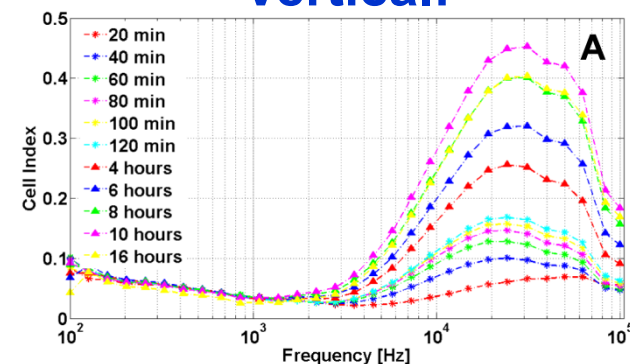
Experimental Results

In PBS, **Good agreement!**

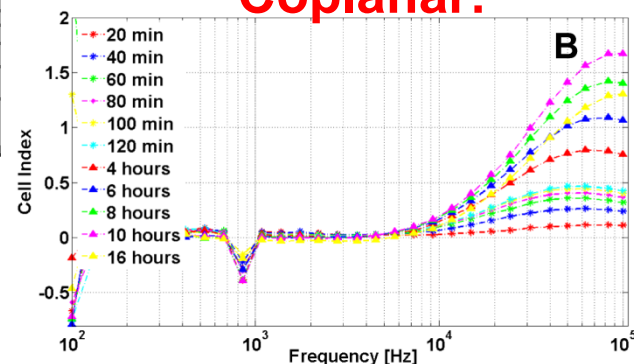


Experimental evidence of f_{max} sensitivity

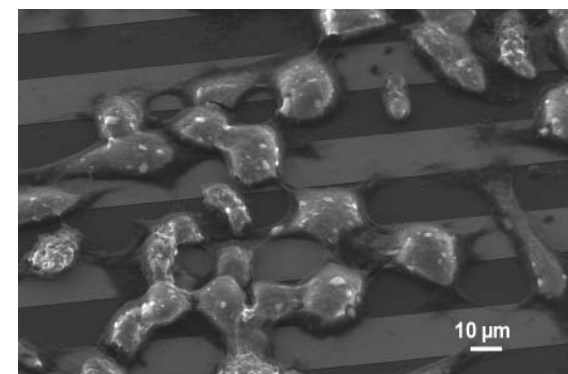
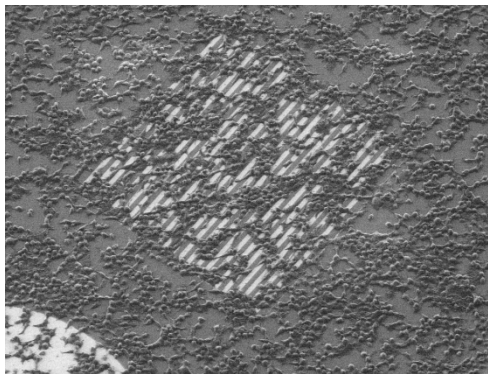
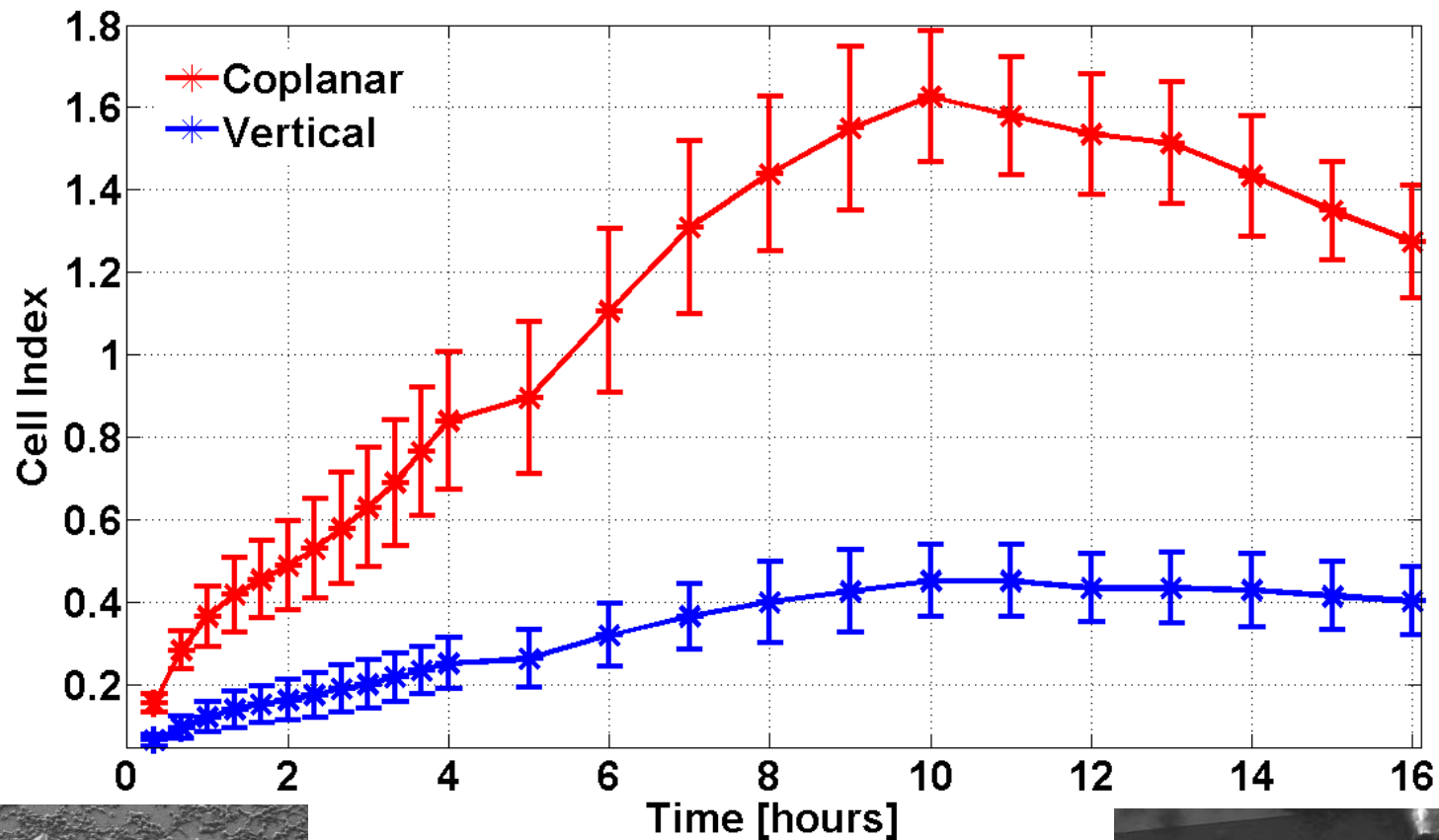
Vertical:



Coplanar:



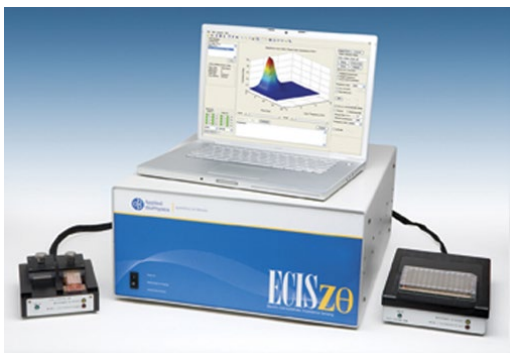
Comparison with HeLa Cells





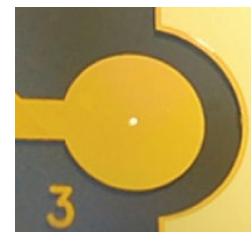
Commercial ECIS Systems

Drug discovery by automatic cyto-toxicity and response analysis:
96-plate wells, 250 μ m – 5mm electrode diameter



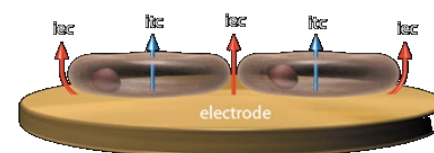
Applied Biophysics
ECIS Z

4kHz - 64kHz

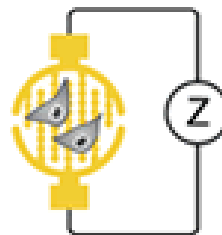


Molecular Devices
Cell Key

1kHz - 10MHz



Roche
exCELLigence



10, 25, 50kHz



- The technique is gaining wide commercial and scientific interest for combinatorial large-scale automatic assays
- The coplanar configuration seems to provide better sensitivity but requires operating at higher frequencies

Limitations:

- Different cell lines have different adhesion properties
- Difficult to obtain a predictive (linear) model
- Difficult to compare quantitative results
- Average population information, no single cell screening